Introduction: -

Electroencephalogram (EEG) was named after Hans Berger, who was the first person to describe oscillations. In 1924, the EEG was named after him. Since then, their roles in several cognitive processes like attention, memory, and perception have received a lot of attention for these brain oscillations. The EEG records electrical activity from the neurons cooperating in neutral coordination within the brains. Such batteries of electrical impulses call for electrodes affixed to the scalp of the subject using what can be considered as metal staples or a head cap supplied with adhesive tabs. Since this method is noninvasive, portable devices appropriate for home use make use of EEG. The parts under observation determine the status of the brain; the later corresponds to the alpha state, delta state, and others depending on the EEG. EEG records variations in brain electrical signals together with organizes them according to wavelengths. Usually, the several bands correspond with certain mental state or function. Gamma waves, for instance, between 30 to 100 Hz are linked to activities requiring concentration, including learning or even brainstorming. It is active when a person is relatively conscious and alert; yet, it is said that an overproduction of beta waves causes stress. Conversely, beta wave frequency decreases during sleep or during periods of meditation. while the person is awake and refreshed, as in rest or sleep or even while the person is day dreaming, alpha rhythm ranging from 8 to 12 Hz is noted. Therefore, tracking and drawing conclusions about the brain in awake, sleep, and throughout various activities depends on the useful measurement of the EEG.

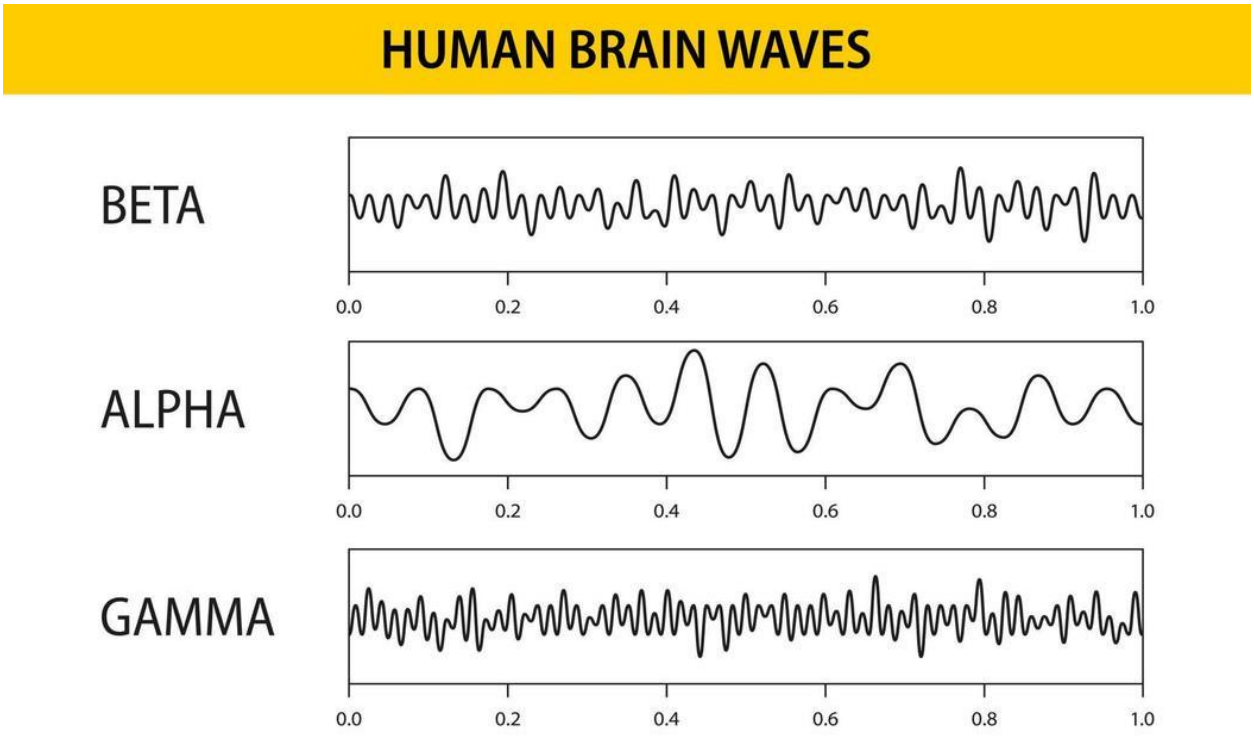


Fig. 1 illustrates these frequency ranges and their Waves.

Table 1: Analyzing several cognitive and physiological situations calls for the different purposes of the Alpha, Beta, Gamma brain waves. Alpha waves are a kind of sleep spindles with an amplitude of 20-200 µ V and varying between 8-12 Hz. It is observed both during meditation when the person is entirely conscious and not nodding off. The frequency range of beta waves spans 13–30 Hz can be classified as low: 13–15 Hz, mid: 16–20 Hz, and high: 21–30 Hz. This has bearing on activity, concentration, even body and mental wakefulness. Included into this state are low beta waves, once referred to as Sensorimotor Rhythm (SMR). Typical mid-beta waves of conscious cognition are high beta waves connected with alert tension. Usually found at the frequencies between 30 and 100 Hz, gamma wave patterns have less than 0. amplitude. This activity is defined by low-amplitude highly frequency oscillations of one microvolt. The link with one of the most evolved areas of the brain is information integration and coordination. These are the kinds of activities that really solve problems and attain the highest level of human performance possible. These influences healing as well as mental capacity enhancement and helps one to clearly see the rest-stimulus-response brain activity.

Table I. Frequency Components of Brain Waves

Brain Wave Type		Frequency Spectrum (Hz)	Amplitude (µV)	Significance
Alpha		8-12 Hz	20-200 µV	• Relaxed, non-sleepy state (calm). • Seen in meditation (restful). • Balances relaxation and alertness (tranquility).
Beta	Low	13-15 Hz	5-20 µV	• Formerly SMR, relaxed yet focused, integrated
	Mid	16-20 Hz		• Thinking, aware of self & surroundings
	High	21-30 Hz		• Alertness, agitation
Gamma		30-100 Hz	<0.1 µV	• High-level cognitive functions (intellectual). • Integrates brain information (synthesis). • High mental activity (peak performance).

System Model Description

Figure 2 shows the analog and digital methods enable to detect EEG signals. The dry EEG electrode captures the electrical activity of the brain first, without requiring conductive gel so that it is better for user to experience and a little bit drier. The instrumentation amplifier amplifies this signal which also small and noisy. A high pass filter eliminates low-frequency noise, such as drift or baseline wandering while preserving the important high-frequency components of your EEG signal. The essence of the brainwave frequencies is preserved, with higher-end noise like potential muscle artefacts attenuated by low pass filtering. The second type of filter is a notch one that removes power line interference at either 50 or 60 Hz, this clears up the signal. These processed EEG signals are subsequently handled by the data processing unit, which converts them from analog to digital form for storage and real-time display provides monitoring. The real-time display provides the feedback to our EEG data instantly, which is essential for applications where one requires a quick response. The EEG data can be transferred remotely with the help of wireless transmission by a specially formed module, making it mobile and comfortable for human use. This data is then streamed to the cloud, which offers much larger storage and computational muscle for deeper analysis as well as remote access. Raw data: the unaltered signals directly from electrodes, useful for subsequent analysis. Utilizing the most advanced post processing algorithms for data analysis, turning raw EEG into interpretable insights. Neural networks are used in AI modeling to identify patterns associated with different states of mind or various neurological disorders. Processed data, models and results are managed adequately on our secure platform for future reutilization to prevent the risk of missing processed dataset. Lastly, the dashboard has a user-friendly interface where users can visualize and interact with EEG data/image analysis results.

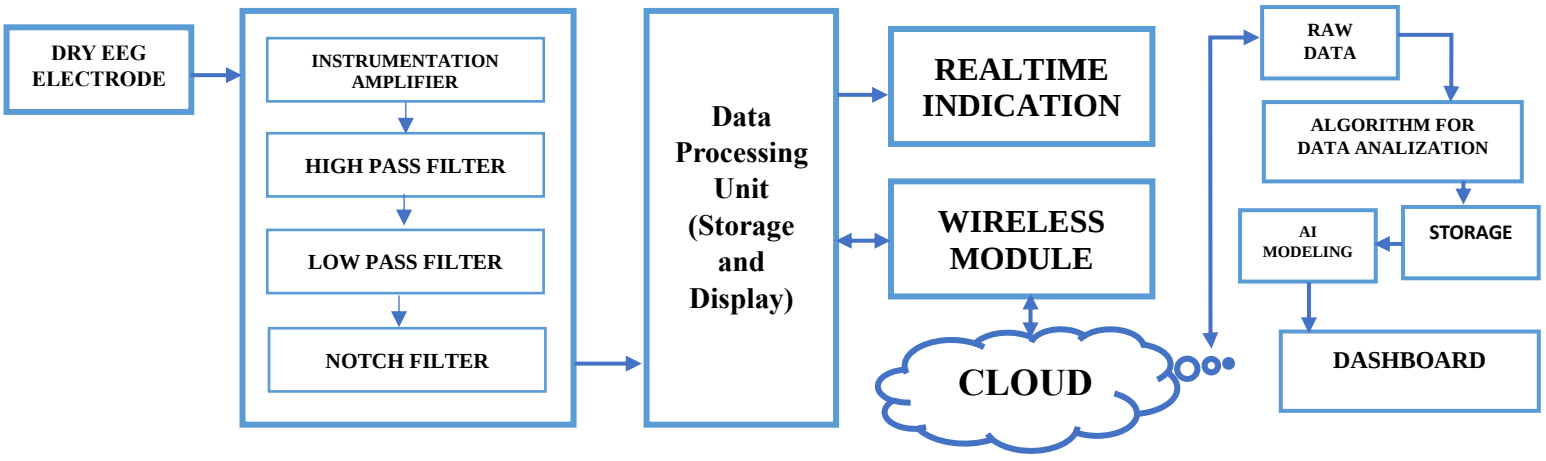


Fig. 2 Block Diagram of EEG Signal Acquisition and Processing System

2.1 EEG Signal Conditioning Unit

The EEG signal conditioning unit is specifically developed to suppress and re-scale the signals so that they are suitable for further DSP analysis. These voltages, typically in the range of 5 to 500 µV and limited to within a frequency band from approximately 1 —100 Hz., are first recorded using electrodes. Signal Acquisition. This has multiple important stages, each playing a part lend making sure the signal is well conditioned. A pair of active high and low-pass filters are used to keep the signal in your intended frequency range, cutting off unwanted frequencies. A notch filter is included to remove the 50 Hz power line interference, greatly improving signal clarity. After filtering stages, a post amplifier is utilized to amplify the signal followed by a voltage adder circuit that ensures electrode nodes only provide brainwave in positive domain. Both the active high-pass and low pass filter circuits have been implemented in a third order Butterworth Filter. The filter is selected for its "maximally flat" frequency response, meaning the passband stays as flat as possible from 0 Hz (DC) all the way up to the cutoff frequency (-3 dB) without any roll-off. Instead, you should enjoy a cleaner EEG directly usable for digital processing. Here The Basic Block Diagram of EEG Signal Acquisition system Is animated in Figure 3. Figure 3 below shows the detailed architecture, which starts with the signal received by electrode then a block of an instrumentation amplifier, active high-pass and low pass filters followed notch filter afterwards post amplified voltage is given to Voltage Adder, a DAQ card so that safe accurate output can be stored or displayed using Data processing unit.

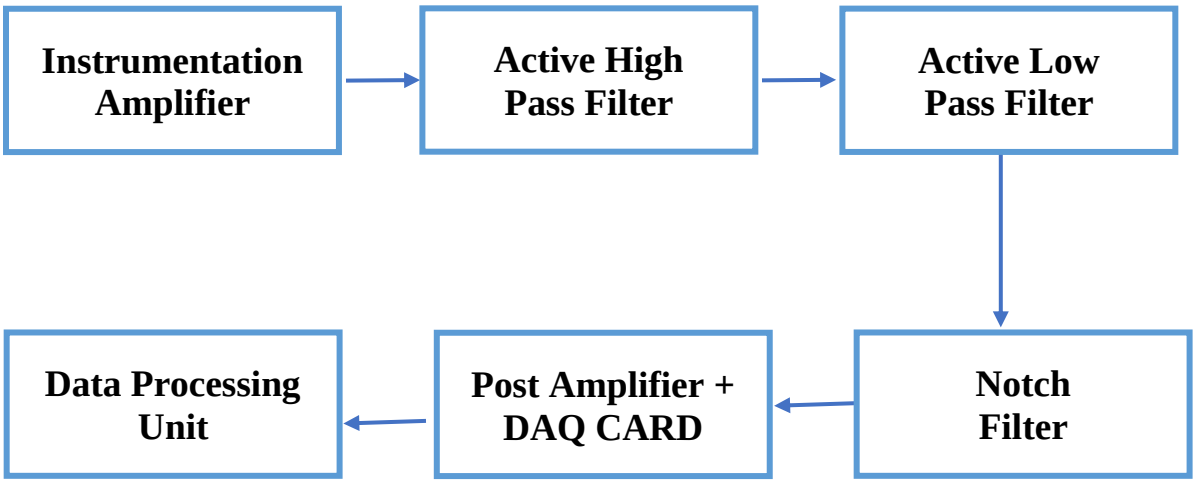


Fig. 3 Block Diagram of EEG Signal Acquisition Framework

The procedure of acquiring the unprocessed EEG signals captured by the electrodes depends on signal conditioning which is the fundamental approaches. This includes a set of operations, which aims at conditioning, amplification, filtering and subsequently preparing signals to go through the other processes.

Instrumentation Amplifier

In particular, the instrumentation amplifier is employed to amplify signal of bio-potentials which are initially observed from the EEG electrodes are small in magnitude. These signals are normally on the order of the microvolt range and can be heavily masked with noise and interference depending upon the situation. That's why this amplifier is able to amplify the signals without distorting them due to its features such as having stable power supply, high input impedance and low output impedance. Based on these factors the inverting operational amplifier INA128P Burr-Brown is proposed to use as the planned IA for the acquisition of the EEG signals; the system is also proposed as the front end. The above mentioned PIR sensor is the INA 128P which was used here in this work due to its superior performance revealed in its use in this application as compared to other similar ICs in this application. This is a single supply instrumentation amplifier which incorporates three operational amplifiers and which has special architecture of the amplifiers to provide an accurate, power efficient and all in one amplifier. The INA 128P has several notable characteristics, including a maximum offset voltage of 50 µV, a maximum drift of 0.5 µV/°C, a maximum input bias current of 5 nA, a minimum Common Mode Rejection Ratio (CMRR) of 120 dB, input protection up to 40V, and a supply range of 2.25V to 18V.
By setting the gain of the INA 128P through a 10k Ω POT that takes on the role of the external resistor, the readings that show a total offset of about 30mV can be achieved.

The calculation of the gain can be determined using the following formula: The computation of the gain as it can be indicated using the following formula too below.

$$G = (1 + \frac{50k\Omega}{R_G})$$

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where R_G is the external resistor.

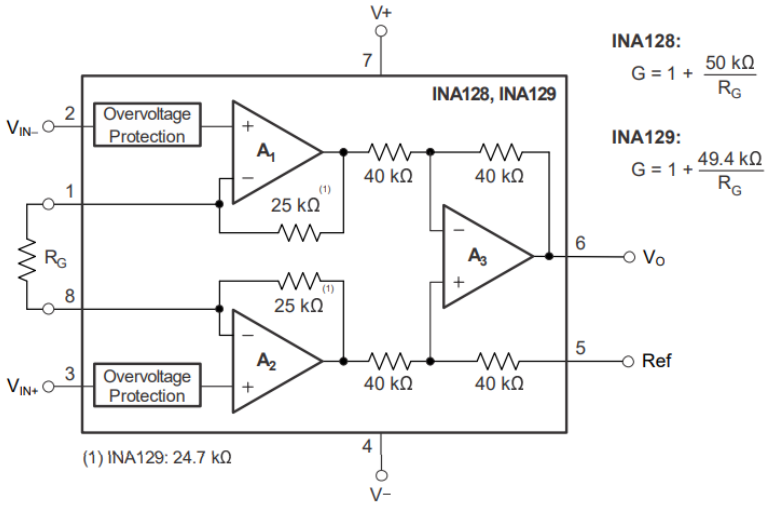


Fig. 3 Instrumentation amplifier INA 128

ACTIVE HIGH PASS FILTER

After the IA, the overall EEG signal is then filtered using what is called the high-pass filter. Since the low-frequency DC component is destructive to the EEG data, this filter is very essential to eliminate it. Thus, eliminating the DC offset, the high-pass filter amplifies the useful signal and excludes interference from the brain's electrical activity. The active high-pass Butterworth filter is developed to counter low-frequency noises especially those that overlap with the raw EEG signal. This filtering process helps cut-off any baseline drifting due to low-frequency noise given that the cut-off frequency of this filter was 1 Hz. Figure 4, presents circuit diagram of the high pass Butterworth filter. The cut off frequency is calculated by: The cut off frequency is calculated by:

$$fc = (\frac{1}{2\pi RC})$$

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Where, fc is the cutoff frequency

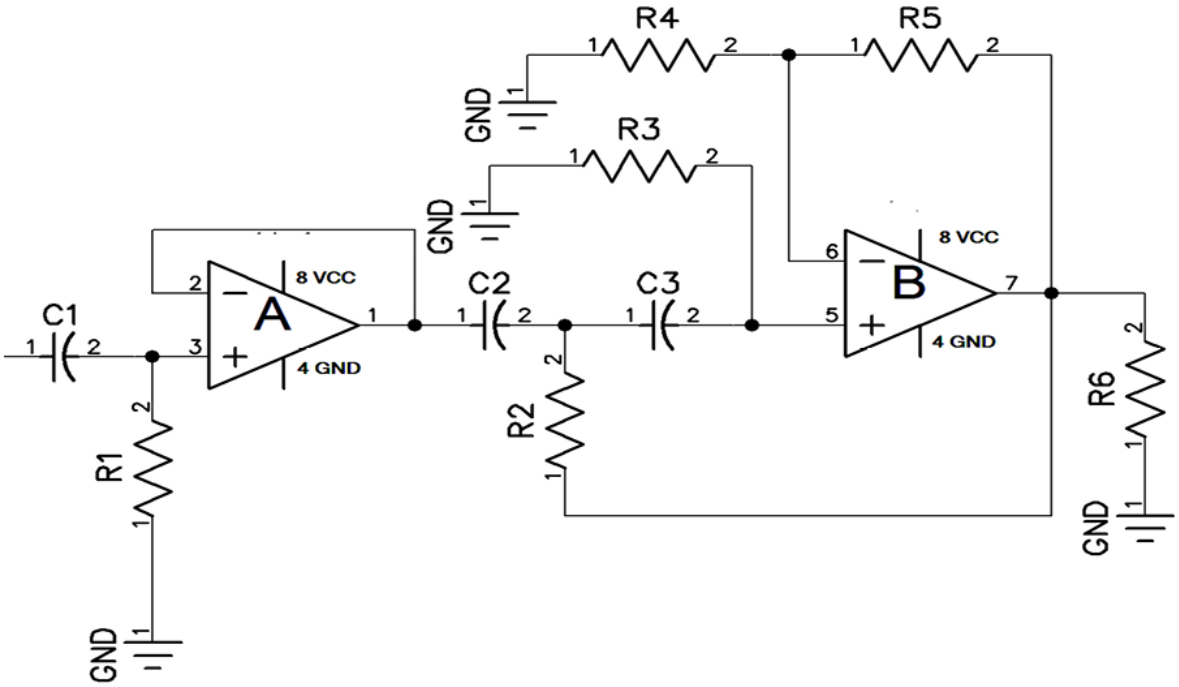


Fig. 4 ACTIVE HIGH PASS FILTER

ACTIVE LOW PASS FILTER

After passing the signal through the high-pass filter, the signal is then passed through a low-pass filter. Due to several sources of electrical interferences and artifacts present in the EEG signals, this stage is important for filtering out of high frequencies noises. The raw EEG that is acquired very often contains unwanted high frequency components and for this reason; application of a low pass filter is inevitable. In the designed active low-pass filter the cut off frequency is 100 Hz, this means that any signals with frequency above 100 Hz are attenuated. The cut-off frequency (fc) of the active low-pass Butterworth filter is calculated using the formula: The cut-off frequency (fc) of the active low-pass Butterworth filter is calculated using the formula:

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The pathway of the EEG signal conditioning is also shown in the circuit diagram of the active low-pass Butterworth filter, which is depicted in Figure 5.

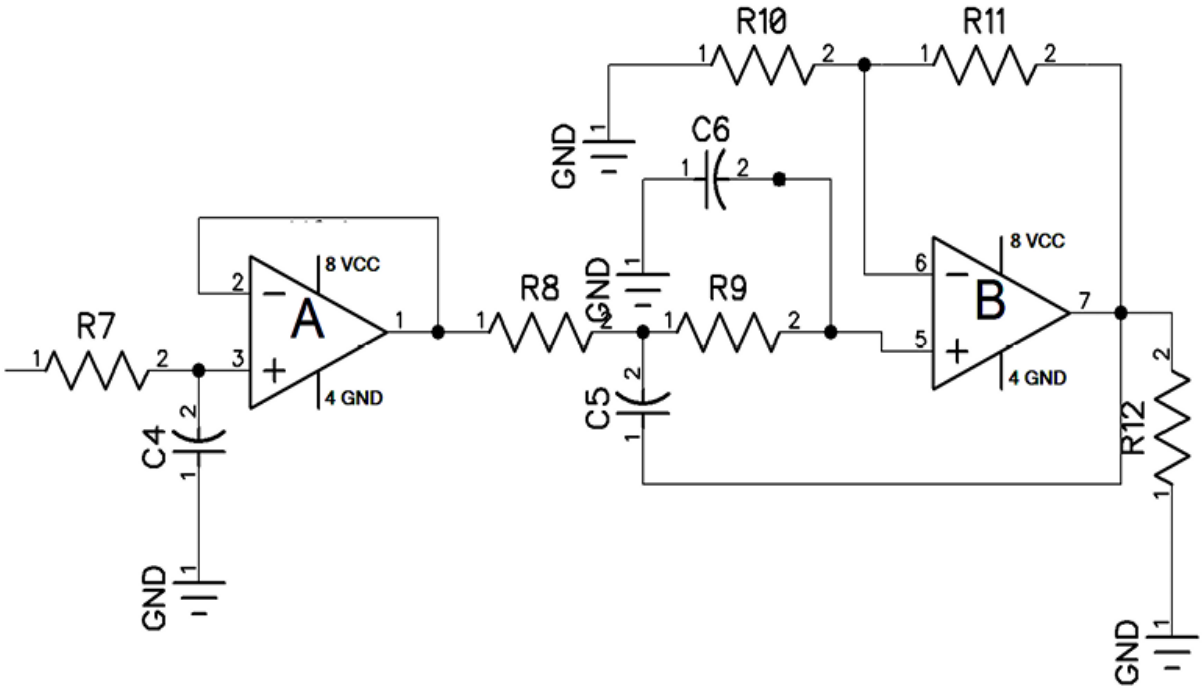


Fig. 5 ACTIVE LOW PASS FILTER

NOTCH FILTER

Following passing through the low-pass filter, the EEG signal is led to the notch filter, which plays an essential role in removing powerline interference usually at 50Hzor 60Hz according to the area. A major source of noise when recording EEG is powerline interference especially where there is closely associated electrical equipment or power lines. The notch filter is on the other hand developed to selectively remove this interference while preserving the desire EEG signal. Although there are issues working to achieve a perfect 50 Hz hardware notch filter – primarily due to issues in achieving precise resistances and capacitance – it stands as a very necessary factor when seeking to gain clarity from the EEG data gathered. The notch filter also keeps the 50 Hz noise away from the EEG signal while leaving most of the other frequencies, which could be useful for subsequent analysis and interpretation. This step is especially crucial when one aims at obtaining reliable EEG data that does not have the typical electrical interference.

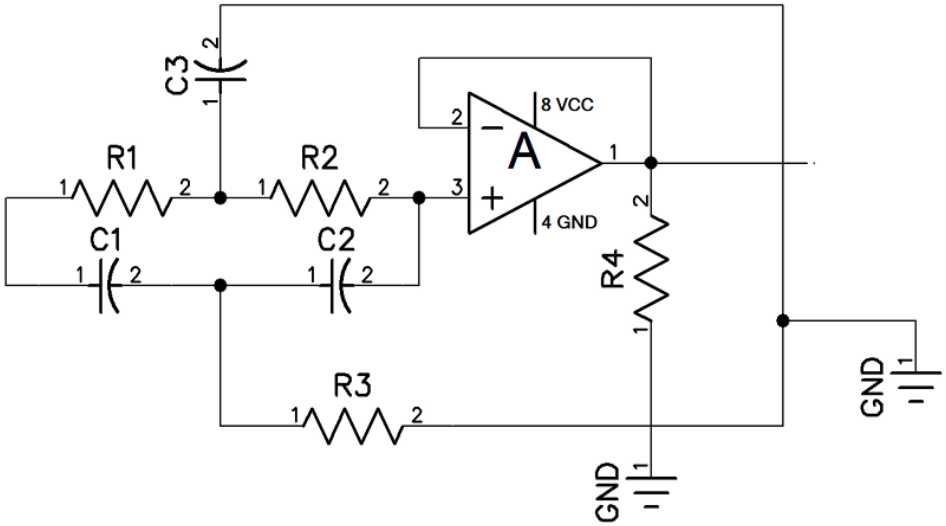


Fig. 6 Notch Filter

Post Amplifier + DAQ Card:-

That after passing through the notch filter the EEG signal goes to the post amplifier stage. It raises the gains on the EEG signal to a level that it has enough amplitude and hence becomes easy to digitize and analyze. Usually, in this case, a non-inverting amplifier configuration is preferred because it has a high input impedance and stable gain. After amplification, the signal is led to an adder voltage circuit that converts the complete signal in the positive voltage system, which is critical for digitizing. This is done with a help of a voltage divider circuit made of resistors and giving an output voltage offset. The amplification can range from 1 to 1000 times the filtered EEG signal, calculated using the formula:

$$G = (1 + \frac{R_f}{R_{in}})$$

G=(1+\frac{R_{f}}{R_{in}})

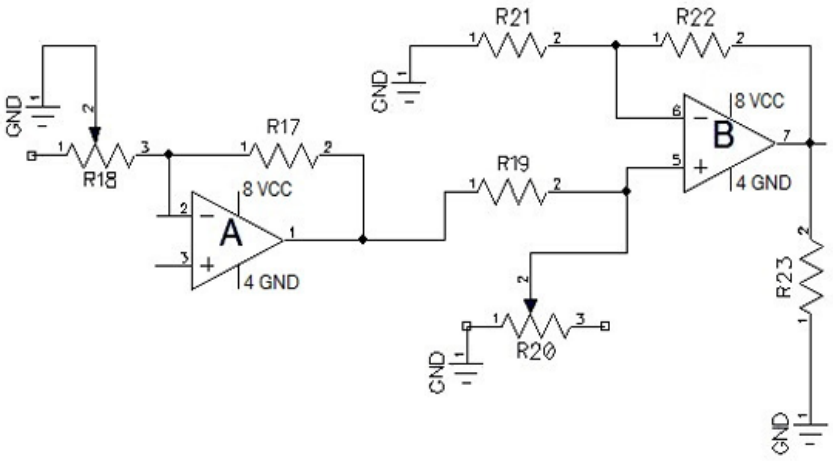


Fig. 7 Post Filter and Voltage adder circuit

The Data Acquisition (DAQ) card can also be described as being central in the EEG signal processing system because it converts the analog EEG signals into digital form. These are then translated by the DAQ card to digital form that can be understood by the computer. To be able to capture all the information the amplified and filtered EEG signals it samples at a high rate. Due to the mentioned factors, the EEG signal undergoes a significant number of modification processes in the amplification, filtering and digitization stages so that it is fit for characterization on the computer.

The DAQ card then analog to digital converts the amplified and filtered EEG signal so that the data can be analyzed on a computer. The digital EEG signals are then processed and temporarily stored, prepared, and initially analyzed by the data processing unit. This consists of converting the analog signals to digital, stashing the data to later on for use and giving actual time output to check on the quality of the recording as well as the placement of the electrodes. The component of real-time indication provides quick impulses on the EEG data, such as pictures of the patterns of the brainwaves and signals, along with indicators that report abnormalities for real-time observation, crucial in using neurofeedback or BCI. The EEG data is transmitted wirelessly to other systems and this enables the subject to be mobile and comfortable particularly when used at home, for the mobile monitoring and clinical purposes. Data is then uploaded to cloud, making use of storage and compute capabilities and optionally applying complex analyses. It allows operation from a distance, work with others in the framework of analysis, and the integration of other services. The raw data is data collected directly from the electrodes which is not yet processed and is stored for data analysis or the same data reprocessed with other algorithms at a later time using other methods or even filters. Data analysis is an algorithm of computed processes on the raw EEG data and applies signal processing, machine learning models and statistical tests for the extraction of useful information, anomalies detection and support of the different application in neuroscience, psychology, and medicine. AI modeling involves utilizing various artificial intelligence approaches in the analysis of EEG signals, recognizes certain patterns called mental states or a certain cognitive task or a neurological disorder. Last but not least, there is a dashboard that enables the user to peruse and navigating the EEG data and results as well as real-brain-wave activity's summary and future prediction; they provide an easily accessible way for the clinician or the researcher or user to understand and use the information.